Lecture 15: Quadratic Penalty and Augmented Lagrangian Methods

MA3236 Non-Linear Programming

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S17-08-21

Quick review

- General constrained problems:

$$\text{local minimum} + \text{LICQ} \implies \text{KKT FONC and SONC};$$

$$\text{KKT FONC and KKT SOSC} \implies \text{strict local minimum}.$$

- Canonical convex problems: at a feasible point satisfying LICQ:

$$\text{KKT FONC} \iff \text{global minimum}.$$

Four components of the KKT violation

Algorithms usually target an **approximate KKT point**.

For $\mathbf{g}(\mathbf{x}) = \mathbf{0}$, $\mathbf{h}(\mathbf{x}) \leq \mathbf{0}$, let

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^T \mathbf{h}(\mathbf{x}),$$

$$\begin{aligned}
&\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = \mathbf{0}, \\
&\mathbf{g}(\mathbf{x}) = \mathbf{0}, \quad \mathbf{h}(\mathbf{x}) \leq \mathbf{0}, \\
&\boldsymbol{\lambda} \in \mathbf{R}^m, \quad \boldsymbol{\nu} \in \mathbf{R}^p_+, \\
&\nu_j h_j(\mathbf{x}) = 0, \quad j = 1, \ldots, p,
\end{aligned}$$

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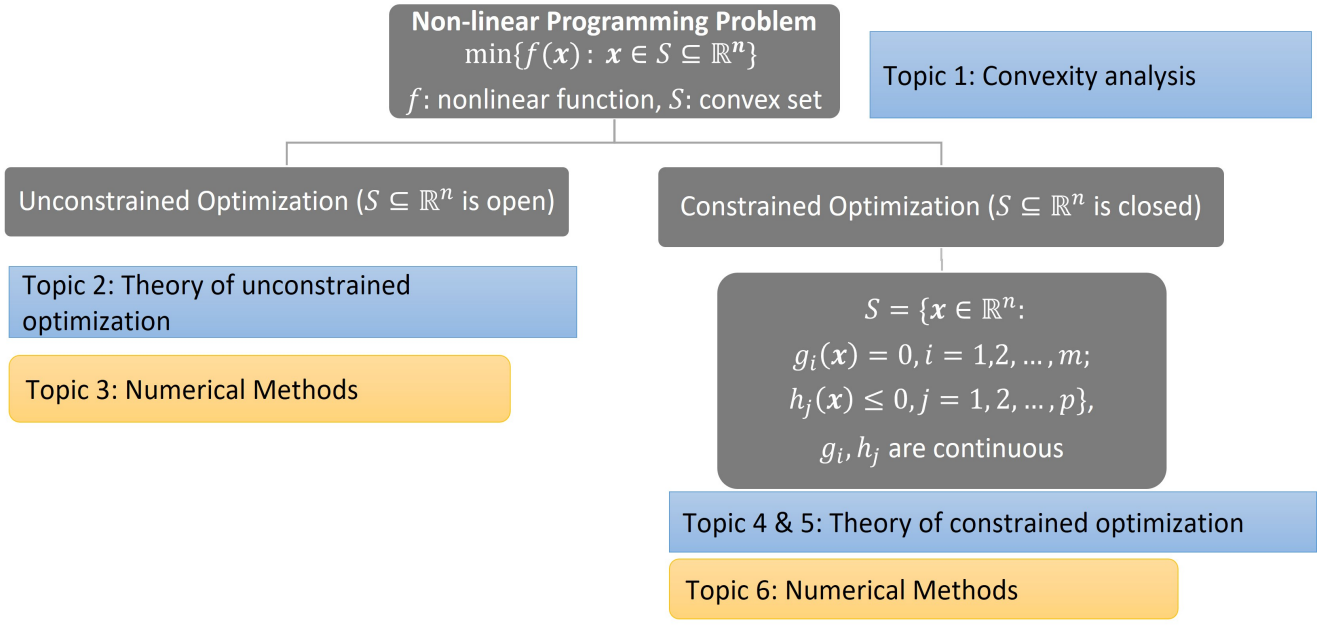
$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^T \mathbf{h}(\mathbf{x}), \qquad [\mathbf{w}]_+ := (\max\{w_i, 0\})_i.$$

$$\begin{aligned}
&\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = \mathbf{0}, && r_{\text{stat}} = \|\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})\|_\infty, \\
&\mathbf{g}(\mathbf{x}) = \mathbf{0}, \quad \mathbf{h}(\mathbf{x}) \le \mathbf{0}, && r_{\text{feas}} = \max\{\|\mathbf{g}(\mathbf{x})\|_\infty, \|[\mathbf{h}(\mathbf{x})]_+\|_\infty\}, \\
&\boldsymbol{\lambda} \in \mathbf{R}^m, \quad \boldsymbol{\nu} \in \mathbf{R}^p_+, && r_{\text{dual}} = \|[-\boldsymbol{\nu}]_+\|_\infty, \\
&\nu_j h_j(\mathbf{x}) = 0, \quad j = 1, \ldots, p, && r_{\text{comp}} = \max_{1 \le j \le p} |\nu_j h_j(\mathbf{x})|.
\end{aligned}$$

$$r_{\text{KKT}} = \max\{r_{\text{stat}}, r_{\text{feas}}, r_{\text{dual}}, r_{\text{comp}}\}.$$

Warning. Small KKT violations alone do not certify a minimum.

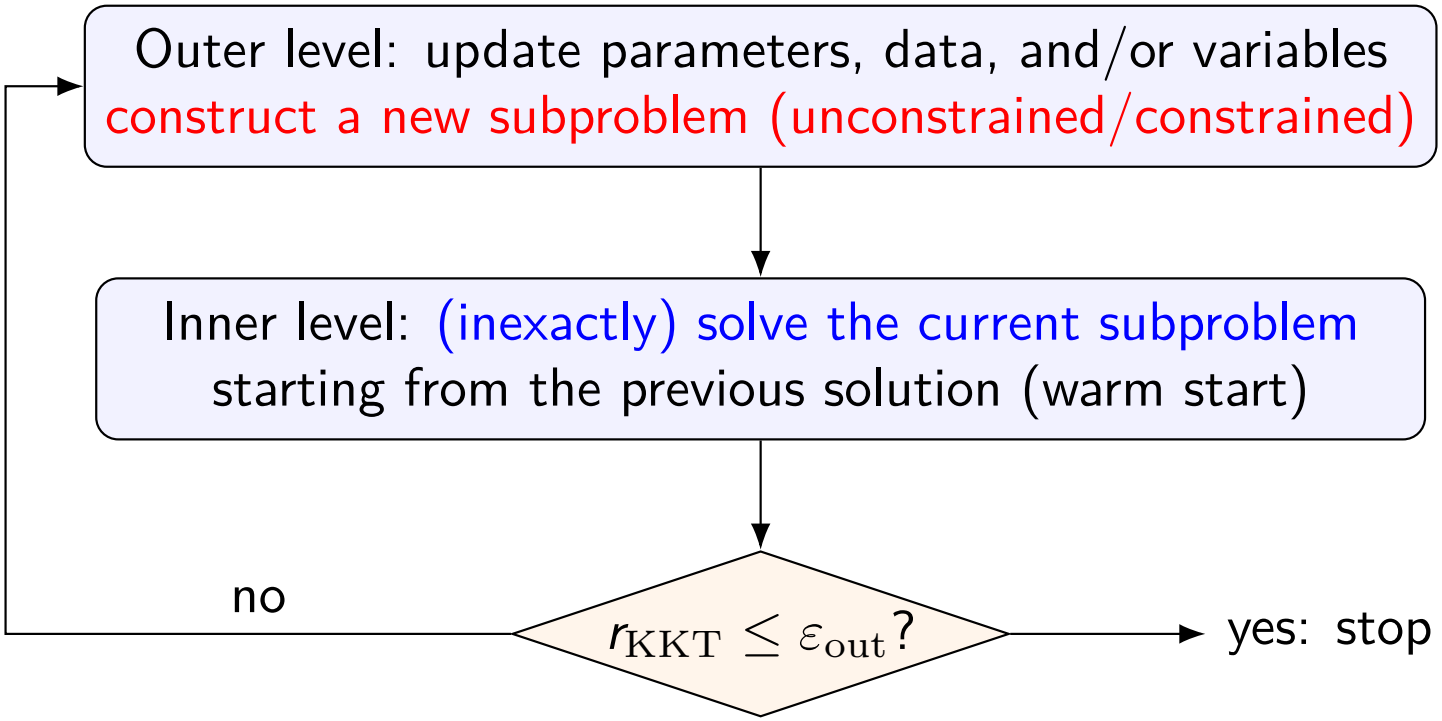
Topics



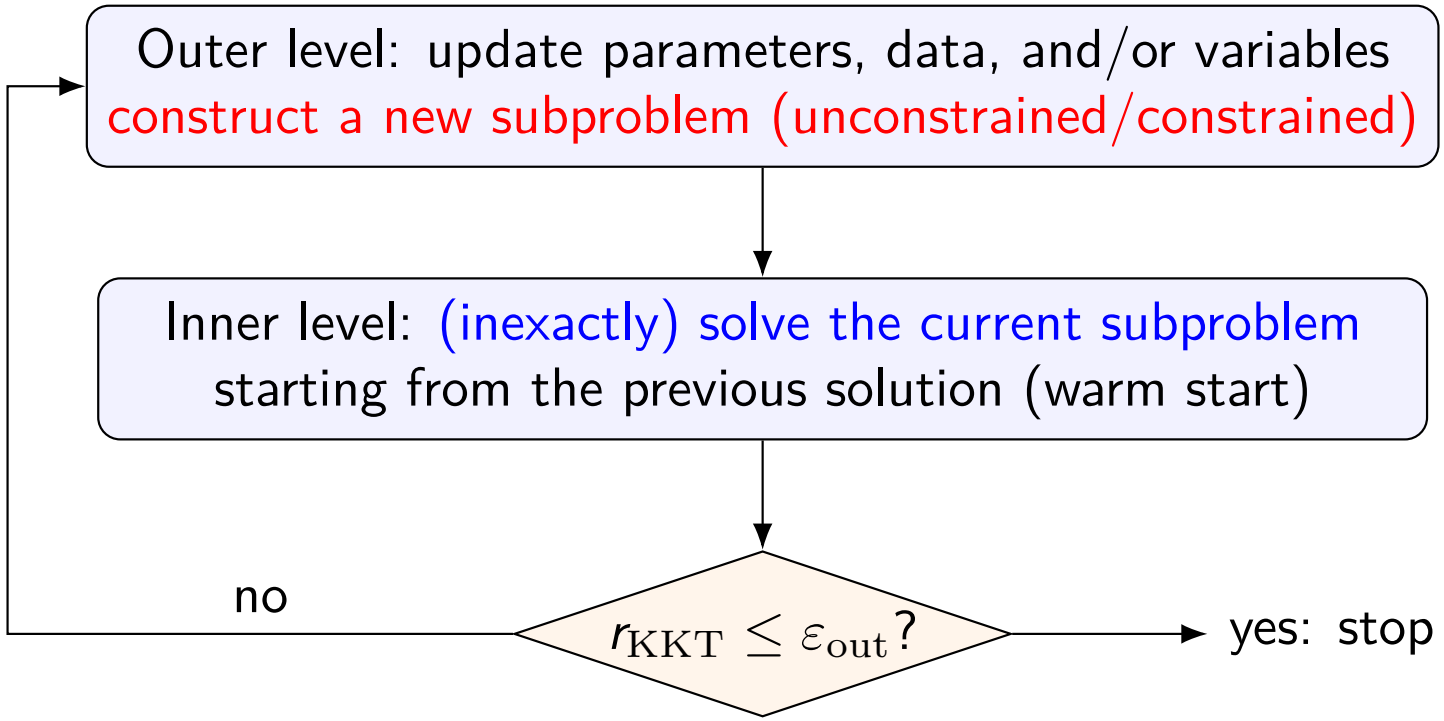
In Topic 6, we study numerical methods for solving **constrained optimization problems**.

- Quadratic penalty and augmented Lagrangian methods
- Lagrange–Newton and sequential quadratic programming
- Barrier and primal interior-point methods
- Subgradient methods for the dual problem (after duality theory)

Two-level iterative framework (in this course)



Two-level iterative framework (in this course)



This lecture: **primal methods with unconstrained subproblems**.

Equality-constrained problems

Assume $f, g_1, \ldots, g_m$ are smooth on $\mathcal{D}$, with $\mathcal{D} \subseteq \mathbf{R}^n$ open:

$$\min_{\mathbf{x} \in \mathcal{D}} f(\mathbf{x}) \qquad \text{s. t.} \qquad \mathbf{g}(\mathbf{x}) = \mathbf{0}.$$

$$\mathbf{A}(\mathbf{x}) = [\nabla g_1(\mathbf{x}), \ldots, \nabla g_m(\mathbf{x})], \qquad L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}).$$

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Only two KKT components remain:

$$r_{\text{stat}}(\mathbf{x}, \boldsymbol{\lambda}) = \|\nabla f(\mathbf{x}) + \mathbf{A}(\mathbf{x})\boldsymbol{\lambda}\|_\infty, \qquad r_{\text{feas}}(\mathbf{x}) = \|\mathbf{g}(\mathbf{x})\|_\infty.$$

$$\boxed{r_{\text{KKT}} = \max\{r_{\text{stat}}, r_{\text{feas}}\}.}$$

Quadratic penalty

The quadratic penalty method (QP) is also called the simple or Courant penalty method.

$$P_\rho(\mathbf{x}) = f(\mathbf{x}) + \frac{\rho}{2}\|\mathbf{g}(\mathbf{x})\|_2^2, \qquad \rho > 0 \text{ penalty parameter.}$$

- Intuition: larger $\rho \Rightarrow$ a larger price for violating the constraints.
- On the feasible set: $P_\rho = f$.
- Off the feasible set: $P_\rho > f$ when $\mathbf{g} \neq \mathbf{0}$.
- Idea: **solve a sequence of unconstrained problems with** $\rho \uparrow$.

A common test problem

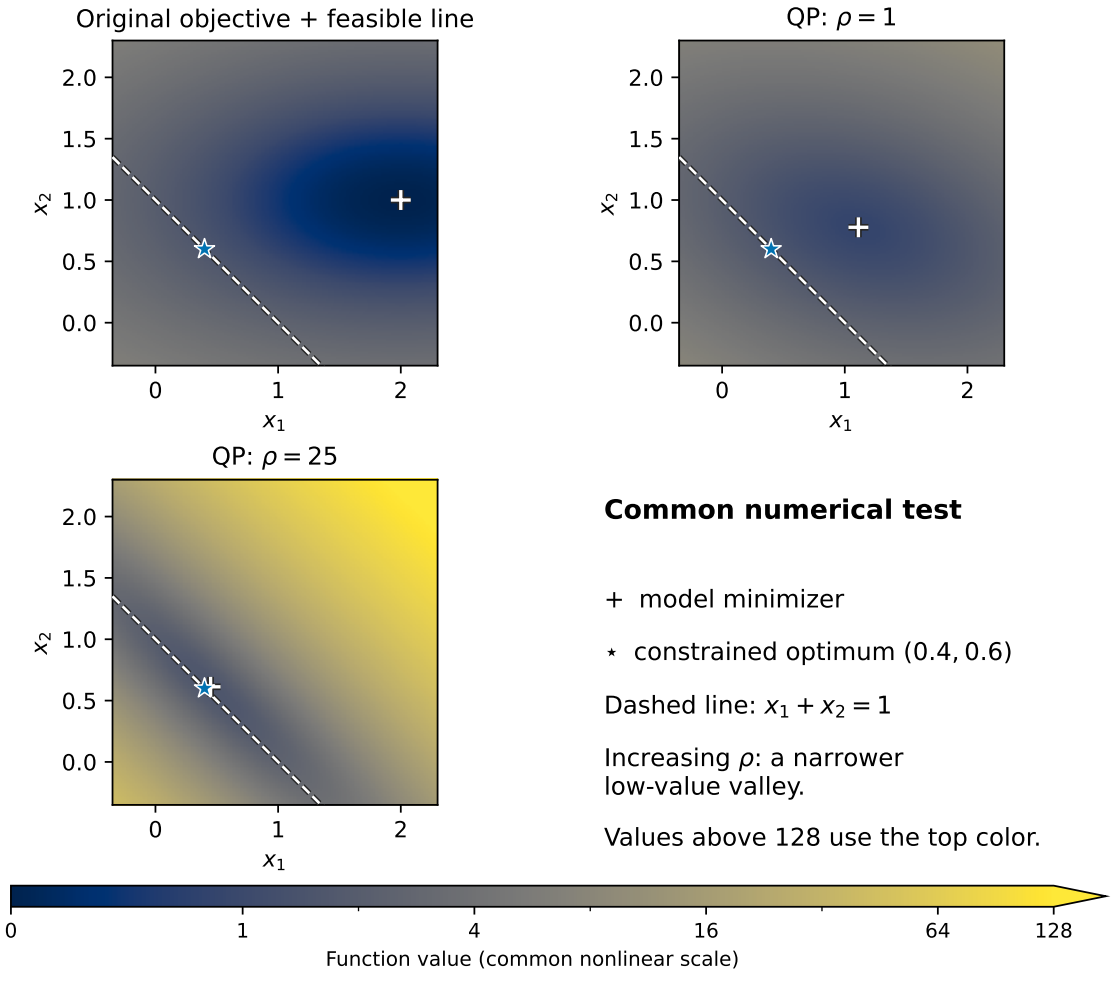
$$\min_{\mathbf{x}\in\mathbf{R}^2} f(\mathbf{x}) = \frac{1}{2}\left[(x_1-2)^2 + 4(x_2-1)^2\right] \qquad \text{s. t.} \qquad g(\mathbf{x}) = x_1 + x_2 - 1 = 0.$$

- Unconstrained minimizer: $(2,1)^T$, which is infeasible.
- The equality KKT system is

$$x_1 - 2 + \lambda = 0, \quad 4(x_2-1) + \lambda = 0, \quad x_1 + x_2 = 1.$$

$$\boxed{\mathbf{x}^* = (2/5, 3/5)^T, \qquad \lambda^* = 8/5, \qquad f^* = 8/5.}$$

QP geometry: a steeper wall around the feasible line



- Increasing ρ steepens the model across $g = 0$.
- The model minimizer approaches the constrained minimizer.

From QP to KKT: a hidden multiplier estimate

$$P_\rho(\mathbf{x}) = f(\mathbf{x}) + \frac{\rho}{2}\|\mathbf{g}(\mathbf{x})\|_2^2 \quad \Rightarrow \quad \nabla P_\rho(\mathbf{x}) = \nabla f(\mathbf{x}) + \rho \mathbf{A}(\mathbf{x})\mathbf{g}(\mathbf{x}).$$

After the k-th subproblem, define

$$\boxed{\boldsymbol{\lambda}^{(k+1)} = \rho^{(k)}\mathbf{g}(\mathbf{x}^{(k+1)}).}$$

From QP to KKT: a hidden multiplier estimate

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After the k-th subproblem, define

$$\boxed{\boldsymbol{\lambda}^{(k+1)} = \rho^{(k)}\mathbf{g}(\mathbf{x}^{(k+1)}).}$$

Then

$$\nabla P_{\rho^{(k)}}(\mathbf{x}^{(k+1)}) = \nabla_{\mathbf{x}} L(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)}).$$

- Exact interior minimization gives $r_{\text{stat}} = 0$.
- An inner gradient tolerance controls r_{stat}.
- The outer loop must still reduce r_{feas}.

From QP to KKT: increasing ρ improves feasibility

Suppose $\mathbf{x}_\rho$ is a global minimizer of P_ρ. For any feasible $\mathbf{y}$,

$$f(\mathbf{x}_\rho) + \frac{\rho}{2}\|\mathbf{g}(\mathbf{x}_\rho)\|_2^2 \leq P_\rho(\mathbf{y}) = f(\mathbf{y}).$$

If $f(\mathbf{x}_\rho) \geq M$ for any ρ, then

$$\boxed{\|\mathbf{g}(\mathbf{x}_\rho)\|_2 \leq \sqrt{\frac{2(f(\mathbf{y}) - M)}{\rho}} \longrightarrow 0.}$$

- For exact global solves, $\|\mathbf{g}(\mathbf{x}_\rho)\|_2$ is nonincreasing in ρ.
- Primal cluster points lying in $\mathcal{D}$ are feasible and globally optimal.

QP: pseudocode

1. Choose $\mathbf{x}^{(0)},\ \rho^{(0)} > 0,\ \gamma > 1$, and fixed tolerances $\varepsilon_{\text{in}}^{(k)} \equiv \varepsilon_{\text{in}}^{(0)} > 0,\ \varepsilon_{\text{out}} > 0$.
2. For $k = 0, 1, \ldots$:
    1. (Approximately) minimize

$$P_{\rho^{(k)}}(\mathbf{x}), \quad \text{such that} \quad \|\nabla P_{\rho^{(k)}}(\mathbf{x}^{(k+1)})\|_\infty \leq \varepsilon_{\text{in}}^{(k)}.$$

    2. Set $\boldsymbol{\lambda}^{(k+1)} = \rho^{(k)}\mathbf{g}(\mathbf{x}^{(k+1)})$.
    3. Stop if $r_{\text{KKT}}(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)}) \leq \varepsilon_{\text{out}}$.
    4. Set $\rho^{(k+1)} = \gamma\rho^{(k)}$, using the same γ every time.

Python code: `quadratic_penalty(..., solver="sd" or "newton")`

QP: unconstrained subsolvers

Subproblems are not necessarily well-posed, convex, or easy-to-solve.

Usual practice:

- Warm-started unconstrained subsolver.
- Approximate local minimizer/stationary point.

QP: unconstrained subsolvers

Subproblems are not necessarily well-posed, convex, or easy-to-solve.

Usual practice:

- Warm-started unconstrained subsolver.
- Approximate local minimizer/stationary point.

For example, until $\|\nabla P_{\rho^{(k)}}(\mathbf{x}^{(k+1)})\|_\infty \leq \varepsilon_{\text{in}}^{(k)}$,

- Line-search SD:

$$\mathbf{d} = -\nabla P_{\rho^{(k)}}(\mathbf{x}) \quad + \quad \text{backtracking.}$$

- Line-search Newton:

$$(\mathbf{H}_{P_{\rho^{(k)}}}(\mathbf{x}) + \tau \mathbf{I})\mathbf{d} = -\nabla P_{\rho^{(k)}}(\mathbf{x}) \quad + \quad \text{backtracking.}$$

QP: exact local inner minima can remain infeasible

$$f(x) = (x-1)^2, \qquad g(x) = x^3 - 3x + 3 = 0.$$

At $x = 1$, for every $\rho > 0$,

$$P_\rho(x) = (x-1)^2 + \frac{\rho}{2}(x^3 - 3x + 3)^2.$$

$$g(1) = 1, \qquad P'_\rho(1) = 0, \qquad P''_\rho(1) = 2 + 6\rho > 0.$$

- $x = 1$ is a strict local minimum of every penalty subproblem.
- Warm-starting there keeps returning an infeasible point.

Inner stationarity, even local inner optimality, does not by itself ensure outer feasibility or a global solution.

QP: a tremendously large ρ may be required

$$\min_{\mathbf{x}\in\mathbf{R}^2} f(\mathbf{x}) = \frac{1}{2}\left[(x_1-2)^2 + 4(x_2-1)^2\right] \quad \text{s. t.} \quad g(\mathbf{x}) = x_1 + x_2 - 1 = 0.$$

$$\Rightarrow \quad P_\rho(\mathbf{x}) = \frac{1}{2}\left[(x_1-2)^2 + 4(x_2-1)^2\right] + \frac{\rho}{2}(x_1+x_2-1)^2$$

Solving the inner equations gives

$$\mathbf{x}_\rho = \frac{1}{4+5\rho}\begin{pmatrix} 8+2\rho \\ 4+3\rho \end{pmatrix}, \qquad g(\mathbf{x}_\rho) = \frac{8}{4+5\rho}.$$

ρ	x_1	x_2	r_{feas}
1	1.111111	0.777778	0.888889
5	0.620690	0.655172	0.275862
25	0.449612	0.612403	0.062016
125	0.410175	0.602544	0.012719

Feasibility improves, but no finite ρ makes $g(\mathbf{x}_\rho) = 0$.

QP: a tremendously large ρ may be required (cont'd)

- At a feasible point,

$$\nabla P_\rho(\mathbf{x}^*) = \nabla f(\mathbf{x}^*).$$

If $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$, then $\mathbf{x}^*$ cannot minimize P_ρ.

QP: a tremendously large ρ may be required (cont'd)

- At a feasible point,

$$\nabla P_\rho(\mathbf{x}^*) = \nabla f(\mathbf{x}^*).$$

  If $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$, then $\mathbf{x}^*$ cannot minimize P_ρ.

- At a regular solution with $\boldsymbol{\lambda}^* \neq \mathbf{0}$,

$$\boldsymbol{\lambda}^{(k+1)} = \rho^{(k)} \mathbf{g}(\mathbf{x}^{(k+1)}) \longrightarrow \boldsymbol{\lambda}^*$$

  requires $\rho^{(k)} \to \infty$ when $\mathbf{g}(\mathbf{x}^{(k+1)}) \to \mathbf{0}$.

QP: a large ρ can make inner solves difficult

$$\mathbf{H}_{P_\rho} = \mathbf{H}_f + \rho \mathbf{A}\mathbf{A}^T + \rho \sum_{i=1}^{m} g_i \mathbf{H}_{g_i}.$$

Near feasibility, the term $\rho \mathbf{A}\mathbf{A}^T$ increases curvature in $\mathrm{Range}(\mathbf{A})$.

$$\text{Test problem:} \quad \mathbf{H}_{P_\rho} = \begin{pmatrix} 1+\rho & \rho \\ \rho & 4+\rho \end{pmatrix}.$$

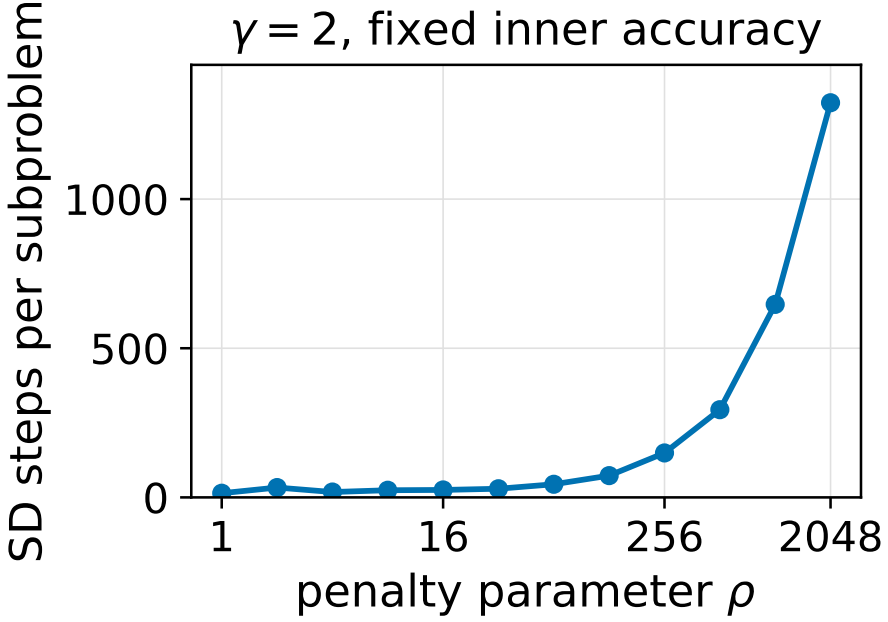
$$\lambda_\pm = \frac{5 + 2\rho \pm \sqrt{9 + 4\rho^2}}{2}, \qquad \kappa(\mathbf{H}_{P_\rho}) \sim \frac{4}{5}\rho.$$

- SD takes many small steps across narrow valleys.
- Newton uses curvature and solves a linear system at each step.

QP: fewer outer iterations can cost more

Same quadratic test: $\mathbf{x}^{(0)} = (2,1)^T$, $\rho^{(0)} = 1$; warm starts.

$$\varepsilon_{\text{in}}^{(k)} \equiv 10^{-4}, \qquad \varepsilon_{\text{out}} = 10^{-3}, \qquad \text{each run uses one fixed } \gamma.$$



γ	Outer solves	Total SD steps	P_ρ evals
2	12	2 673	34 042
5	6	4 998	67 985
10	5	9 755	147 284
20	4	13 841	215 768

- Increasing ρ: more inner SD work, even at fixed accuracy.
- Larger γ: fewer outer solves, but higher total cost here.

Augmented Lagrangian method

QP represents the entire multiplier by

$$\boldsymbol{\lambda}^{(k+1)} = \rho^{(k)}\mathbf{g}(\mathbf{x}^{(k+1)}).$$

The augmented Lagrangian method (ALM) instead uses

$$\boxed{\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho\mathbf{g}(\mathbf{x}^{(k+1)}).}$$

- $\boldsymbol{\lambda}^{(k)}$: a retained estimate of the constraint forces.
- $\rho\mathbf{g}(\mathbf{x}^{(k+1)})$: a correction to this estimate.
- Idea: **small feasibility errors can coexist with a nonzero multiplier without forcing** $\rho \to \infty$.

Subproblem? Motivation?

ALM: subproblem and the multiplier update

$$\boxed{L_A(\mathbf{x}, \boldsymbol{\lambda}; \rho) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \frac{\rho}{2} \|\mathbf{g}(\mathbf{x})\|_2^2.}$$

If an inner point $\mathbf{x}^{(k+1)}$ satisfies

$$\begin{aligned} &\nabla_{\mathbf{x}} L_A(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k)}; \rho) \\ &= \nabla f(\mathbf{x}^{(k+1)}) + \mathbf{A}(\mathbf{x}^{(k+1)}) \left[\boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)})\right] \approx \mathbf{0}, \end{aligned}$$

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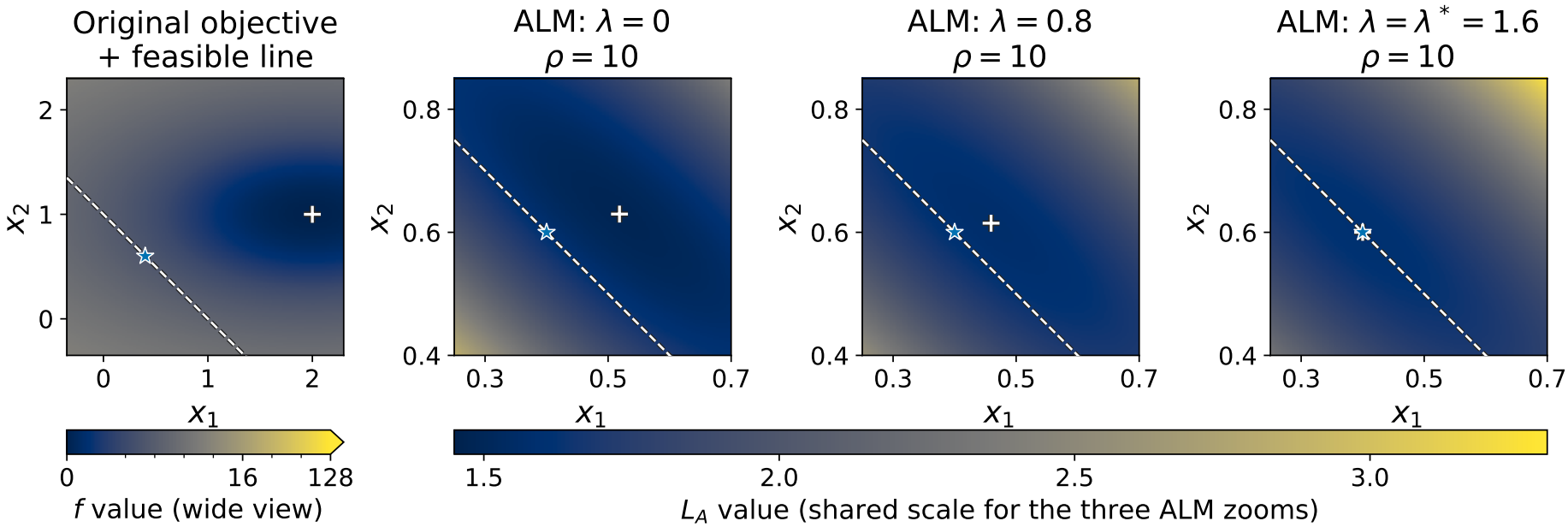
define the updated multiplier

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)}).$$

Then

$$r_{\text{stat}}(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)}) = \|\nabla_{\mathbf{x}} L_A(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k)}; \rho)\|_\infty.$$

ALM: finite penalty with an improving multiplier



ALM panels: the same zoom and value scale, fixed $\rho = 10$.
$+$: inner minimizer; $\star$: $\mathbf{x}^*$; dashed line: $g = 0$.

As λ approaches λ^*, the global inner minimizer approaches $\mathbf{x}^*$ with a fixed ρ;
$\lambda = \lambda^*$ makes $\mathbf{x}^*$ the global inner minimizer for every $\rho > 0$.

ALM: not any positive penalty parameter suffices

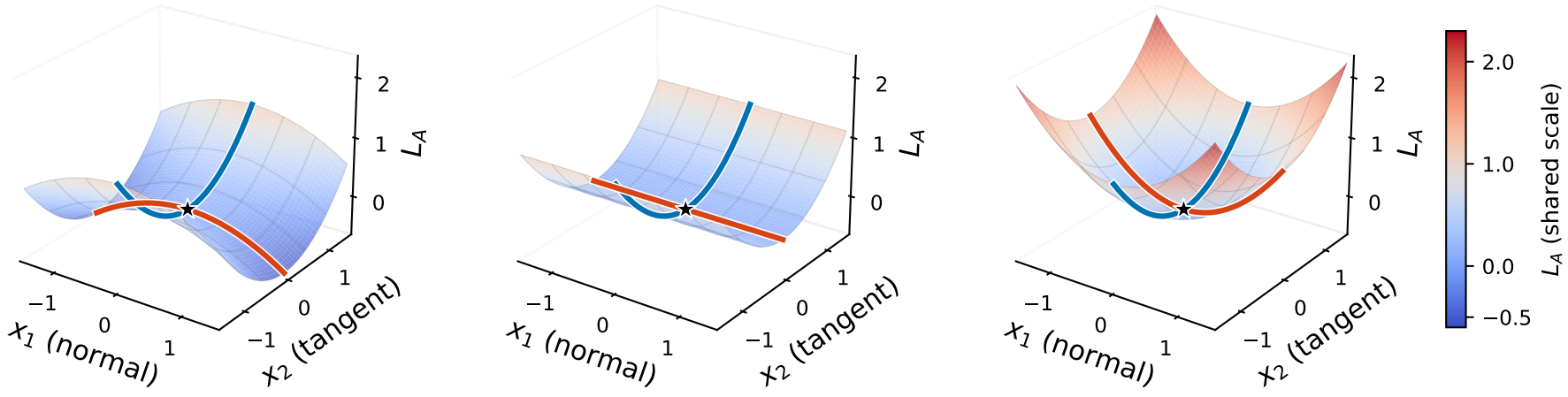
$$\min_{\mathbf{x}\in\mathbf{R}^2} \; -\tfrac{1}{2}x_1^2 + \tfrac{1}{2}x_2^2 \quad \text{s. t.} \quad x_1 = 0, \qquad (\mathbf{x}^*, \lambda^*) = (\mathbf{0}, 0).$$

$$L_A(\mathbf{x}, 0; \rho) = \tfrac{\rho-1}{2}x_1^2 + \tfrac{1}{2}x_2^2, \qquad \mathbf{H}_{L_A} = \operatorname{diag}(\rho - 1, 1).$$

$\rho = 0.5$: saddle

$\rho = 1$: flat valley
(non-strict minimum)

$\rho = 2$: strict inner minimum



feasible slice: $x_1 = 0$, $L_A = x_2^2/2$ normal slice: $x_2 = 0$, $L_A = (\rho - 1)x_1^2/2$

In theory, penalty parameter ρ needs to be large enough!

ALM: pseudocode

1. Choose $\mathbf{x}^{(0)}, \boldsymbol{\lambda}^{(0)}$, a fixed $\rho > 0$, and fixed tolerances $\varepsilon_{\text{in}}^{(k)} \equiv \varepsilon_{\text{in}}^{(0)} > 0,\ \varepsilon_{\text{out}} > 0$.
2. For $k = 0, 1, \ldots$:
   1. (Approximately) minimize

$$L_A(\mathbf{x}, \boldsymbol{\lambda}^{(k)}; \rho), \quad \text{such that} \quad \|\nabla_{\mathbf{x}} L_A(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k)}; \rho)\|_\infty \leq \varepsilon_{\text{in}}^{(k)}.$$

   2. Unconditionally update $\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)})$.
   3. Stop if $r_{\text{KKT}}(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)}) \leq \varepsilon_{\text{out}}$.
   4. Keep ρ and $\varepsilon_{\text{in}}^{(k)}$ fixed for the next subproblem.

Python code: `augmented_lagrangian(..., solver="sd" or "newton")`

The multiplier changes after every inner solve; the penalty stays fixed.

ALM: exact local inner minima can remain infeasible

$$f(x) = (x-1)^2, \qquad g(x) = x^3 - 3x + 3 = 0.$$

For $\lambda \geq 0$ and $\rho > 0$,

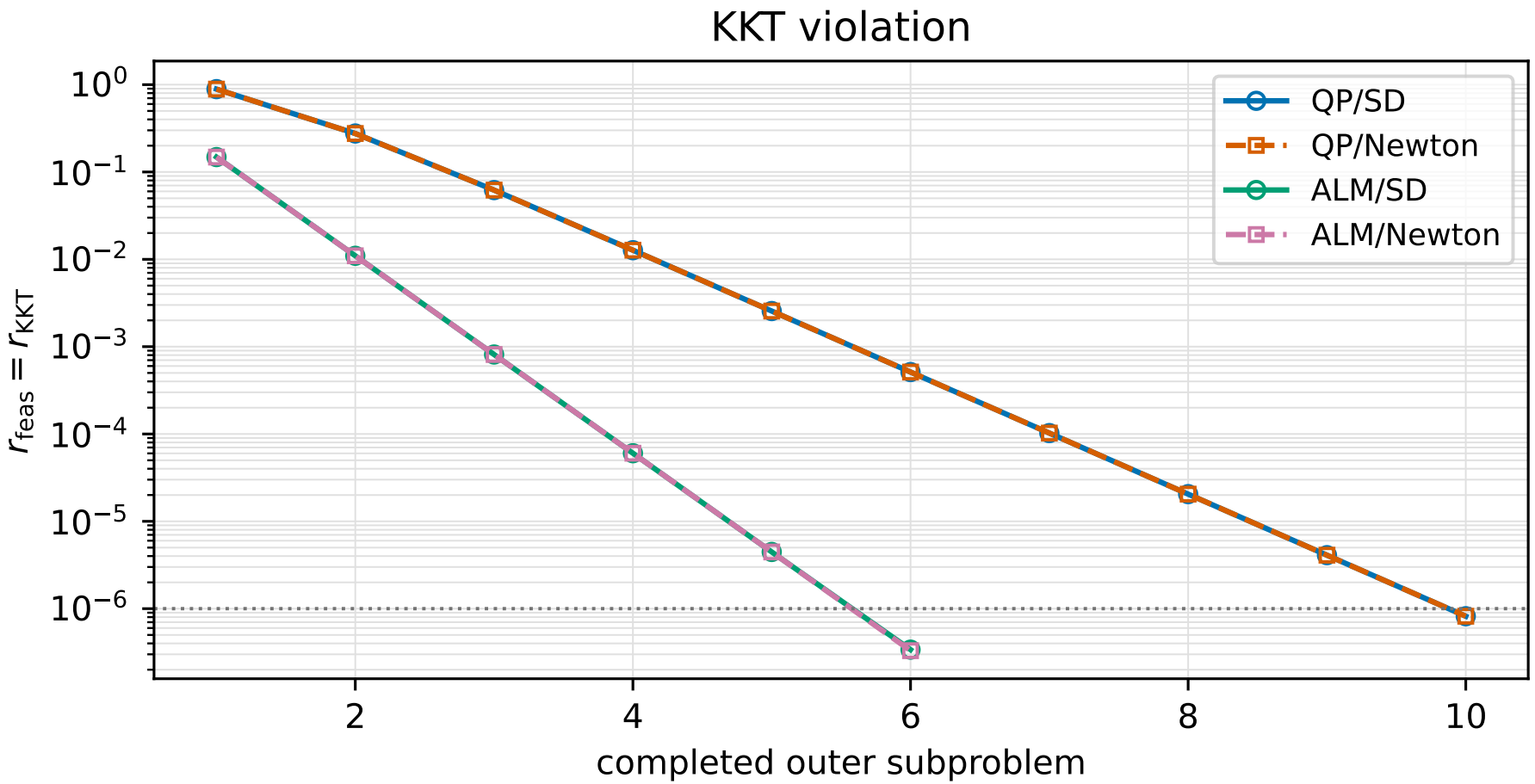
$$L_A(x, \lambda; \rho) = (x-1)^2 + \lambda(x^3 - 3x + 3) + \frac{\rho}{2}(x^3 - 3x + 3)^2.$$

$$g(1) = 1, \qquad L'_A(1) = 0, \qquad L''_A(1) = 2 + 6(\lambda + \rho) > 0.$$

- $x = 1$ is an exact strict local minimum of every inner problem.
- Start at $x^{(0)} = 1$, $\lambda^{(0)} = 0$; warm-started local solves keep $x^{(k)} = 1$, while $\lambda^{(k+1)} = \lambda^{(k)} + \rho$.

For any fixed $\rho > 0$, the multiplier updates cannot escape this infeasible local branch.

QP versus ALM: KKT violation



$\mathbf{x}^{(0)} = (2,1)^T$, $\lambda^{(0)} = 0$; $\varepsilon_{\text{out}} = 10^{-6}$, $\varepsilon_{\text{in}}^{(k)} \equiv 10^{-7}$.
QP: $\rho^{(0)} = 1$, $\gamma = 5$; ALM: fixed $\rho = 10$.

QP versus ALM: more statistics

Method / subsolver	Outer	Inner	Final ρ	Final r_{KKT}
QP / SD	10	3 836 166	1953125	$8.19 \cdot 10^{-7}$
QP / Newton	10	10	1953125	$8.19 \cdot 10^{-7}$
ALM / SD	6	277	10	$3.39 \cdot 10^{-7}$
ALM / Newton	6	6	10	$3.30 \cdot 10^{-7}$

- Here ALM reaches the target with fixed $\rho = 10$.
- It avoids QP's large-penalty conditioning in the late subproblems.
- Newton also solves linear systems; step counts may not be enough.

QP and ALM in practice

Coordinate the following three aspects:

- Desired outer accuracy:
- Inner accuracy:
- Penalty parameters:

with your **budget**!

QP and ALM in practice

Coordinate the following three aspects:

- Desired outer accuracy: choose meaningful, scaled KKT tolerances;
- Inner accuracy:
- Penalty parameters:

with your **budget**!

QP and ALM in practice

Coordinate the following three aspects:

- Desired outer accuracy: choose meaningful, scaled KKT tolerances;
- Inner accuracy: loose solves can leave a stationarity floor; always very tight solves can waste work; local inner minima can remain infeasible;
- Penalty parameters:

with your **budget**!

QP and ALM in practice

Coordinate the following three aspects:

- Desired outer accuracy: choose meaningful, scaled KKT tolerances;
- Inner accuracy: loose solves can leave a stationarity floor; always very tight solves can waste work; local inner minima can remain infeasible;
- Penalty parameters: QP uses a fixed growth factor $\gamma > 1$; larger factors can reduce outer iterations while increasing inner work; ALM uses a fixed $\rho > 0$; its effectiveness depends on the problem;

with your **budget**!

Optimize your optimizer!

Extension: quadratic penalties for inequalities

For

$$\mathbf{g}(\mathbf{x}) = \mathbf{0}, \qquad \mathbf{h}(\mathbf{x}) \leq \mathbf{0},$$

use the positive part only:

$$\boxed{P_\rho(\mathbf{x}) = f(\mathbf{x}) + \frac{\rho}{2}\left(\|\mathbf{g}(\mathbf{x})\|_2^2 + \|[\mathbf{h}(\mathbf{x})]_+\|_2^2\right).}$$

Multiplier estimates:

$$\boldsymbol{\lambda} = \rho\mathbf{g}(\mathbf{x}), \qquad \boldsymbol{\nu} = \rho[\mathbf{h}(\mathbf{x})]_+ \geq \mathbf{0}.$$

- Satisfied inequalities receive no penalty.
- Dual feasibility is automatic for this estimate.
- The inequality models are C^1, but generally not C^2 at switches.

Extension: augmented Lagrangians for inequalities

A Powell–Hestenes–Rockafellar augmented Lagrangian is

$$\boxed{\begin{aligned} L_A(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}; \rho) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \frac{\rho}{2} \|\mathbf{g}(\mathbf{x})\|_2^2 \\ + \frac{1}{2\rho} \left(\|[\boldsymbol{\nu} + \rho \mathbf{h}(\mathbf{x})]_+\|_2^2 - \|\boldsymbol{\nu}\|_2^2 \right). \end{aligned}}$$

After each inner solve, update

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)}), \qquad \boldsymbol{\nu}^{(k+1)} = [\boldsymbol{\nu}^{(k)} + \rho \mathbf{h}(\mathbf{x}^{(k+1)})]_+.$$

- Projection maintains $\boldsymbol{\nu}^{(k+1)} \geq \mathbf{0}$.
- The inequality models are C^1, but generally not C^2 at switches.

Take-home messages

- Small KKT violations do not certify a nonconvex minimum.
- This course: two-level iterative framework.
- This lecture: methods (QP & ALM) for **equality-constrained** optimization with unconstrained subproblems.
- QP: $\boldsymbol{\lambda} \approx \rho \mathbf{g}$; high accuracy often needs large ρ.
- ALM: $\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)})$; finite ρ can suffice.
- Inner solver, tolerances, and penalty updates **mix together**.

Reference: exact QP path on the test problem

The inner equations are

$$x_1 - 2 + \rho(x_1 + x_2 - 1) = 0, \quad 4(x_2 - 1) + \rho(x_1 + x_2 - 1) = 0.$$

Set $\widehat{\lambda}_\rho = \rho(x_1 + x_2 - 1)$. Then

$$x_1 = 2 - \widehat{\lambda}_\rho, \qquad x_2 = 1 - \widehat{\lambda}_\rho/4.$$

Hence

$$\widehat{\lambda}_\rho = \rho\left(2 - \frac{5}{4}\widehat{\lambda}_\rho\right), \qquad \widehat{\lambda}_\rho = \frac{8\rho}{4 + 5\rho}.$$

$$\mathbf{x}_\rho \to (2/5, 3/5)^T, \qquad \widehat{\lambda}_\rho \to 8/5, \qquad g(\mathbf{x}_\rho) = \frac{8}{4 + 5\rho} \to 0.$$

Reference: exact fixed-penalty ALM recurrence

For the current multiplier $\lambda^{(k)}$, inner stationarity gives

$$x_1^{(k+1)} = 2 - \lambda^{(k+1)}, \qquad x_2^{(k+1)} = 1 - \lambda^{(k+1)}/4,$$

where $\lambda^{(k+1)} = \lambda^{(k)} + \rho g(\mathbf{x}^{(k+1)})$. Therefore,

$$\lambda^{(k+1)} = \lambda^{(k)} + \rho \left(2 - \frac{5}{4}\lambda^{(k+1)} \right).$$

Since $\lambda^* = 8/5$,

$$\lambda^{(k+1)} - \lambda^* = \frac{\lambda^{(k)} - \lambda^*}{1 + 5\rho/4}.$$

This is a contraction for every fixed $\rho > 0$.

Reference: the penalty Hessian and conditioning

For the test problem,

$$\mathbf{H}_{P_\rho} = \begin{pmatrix} 1+\rho & \rho \\ \rho & 4+\rho \end{pmatrix}.$$

Its characteristic equation is

$$\lambda^2 - (5+2\rho)\lambda + (4+5\rho) = 0,$$

so

$$\lambda_\pm = \frac{5+2\rho \pm \sqrt{9+4\rho^2}}{2}.$$

As $\rho \to \infty$,

$$\lambda_+ \sim 2\rho, \qquad \lambda_- = \frac{4+5\rho}{\lambda_+} \to \frac{5}{2}, \qquad \kappa = \frac{\lambda_+}{\lambda_-} \sim \frac{4}{5}\rho.$$

Normal curvature grows; tangent curvature stays bounded.

Reference: monotonicity for exact global QP solves

Let $\rho_2 > \rho_1$, and let $\mathbf{x}_i$ globally minimize P_{ρ_i}. Write $v_i = \|\mathbf{g}(\mathbf{x}_i)\|_2^2$ and $f_i = f(\mathbf{x}_i)$.

$$f_1 + \frac{\rho_1}{2} v_1 \le f_2 + \frac{\rho_1}{2} v_2, \qquad f_2 + \frac{\rho_2}{2} v_2 \le f_1 + \frac{\rho_2}{2} v_1.$$

Adding gives

$$(\rho_2 - \rho_1)(v_2 - v_1) \le 0 \quad \Longrightarrow \quad v_2 \le v_1.$$

The first inequality now yields

$$f_2 - f_1 \ge \frac{\rho_1}{2}(v_1 - v_2) \ge 0.$$

- Feasibility improves while the original objective may increase.
- These conclusions do not apply to arbitrary local or inexact solves.

Reference: convex ALM with a fixed penalty

Assume convex f, affine $\mathbf{g}$, an existing KKT pair $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$, and exact global inner solves.

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)}).$$

Convexity and inner stationarity imply

$$(\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^*)^T \mathbf{g}(\mathbf{x}^{(k+1)}) \leq 0.$$

Consequently,

$$\|\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^*\|_2^2 + \rho^2 \|\mathbf{g}(\mathbf{x}^{(k+1)})\|_2^2 \leq \|\boldsymbol{\lambda}^{(k)} - \boldsymbol{\lambda}^*\|_2^2.$$

- Summation gives $\mathbf{g}(\mathbf{x}^{(k+1)}) \to \mathbf{0}$, with fixed $\rho > 0$.
- A primal cluster point in $\mathcal{D}$ is a KKT point, hence a global solution.
- This does not ensure a primal cluster point exists.